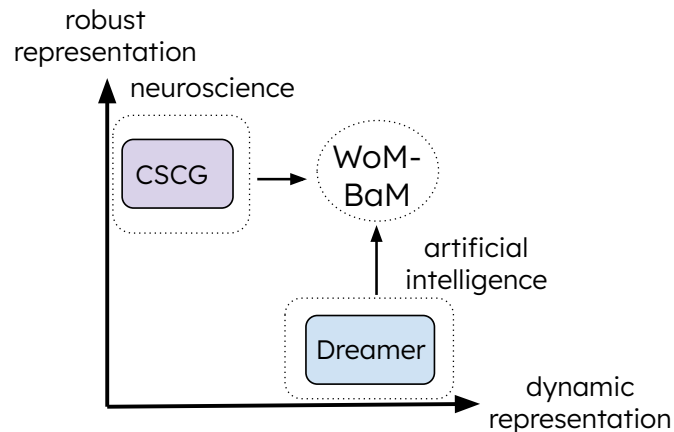
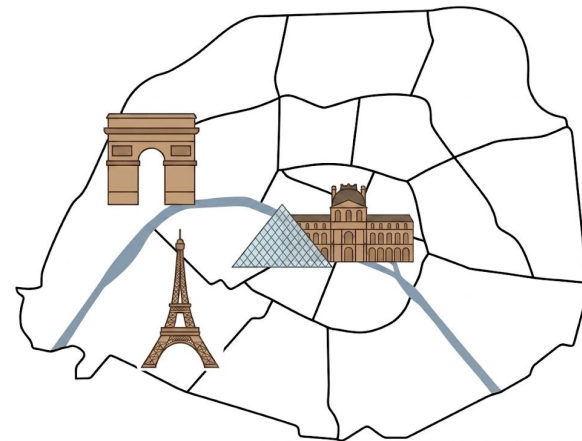


WoM-BaM

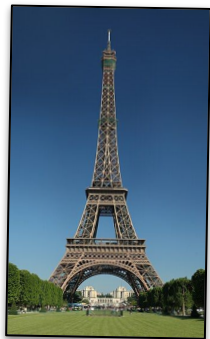
World Models in Brains and Machines

From
Normative Hippocampus Models
to
Robust Planning in Artificial Intelligence
and back

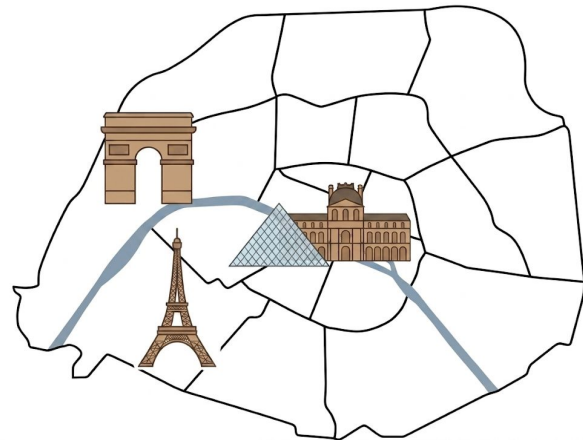
Tristan Manfred Stöber



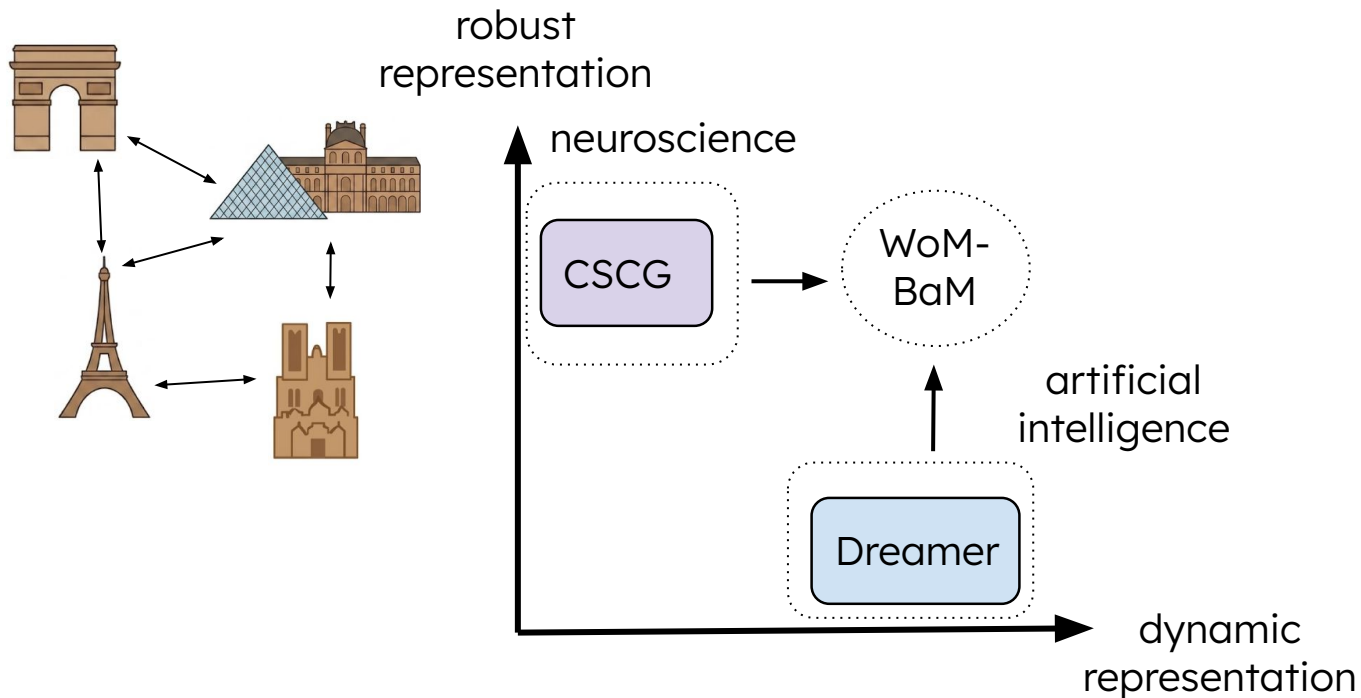
A marvellous feat of our brain: Extracting structure from sequential observation



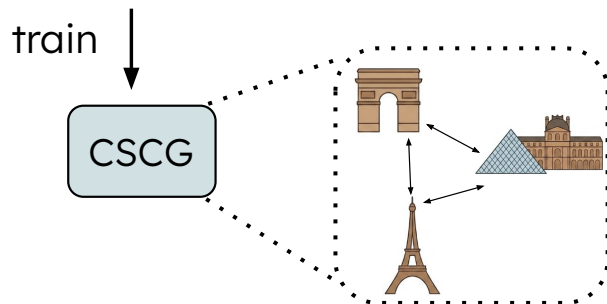
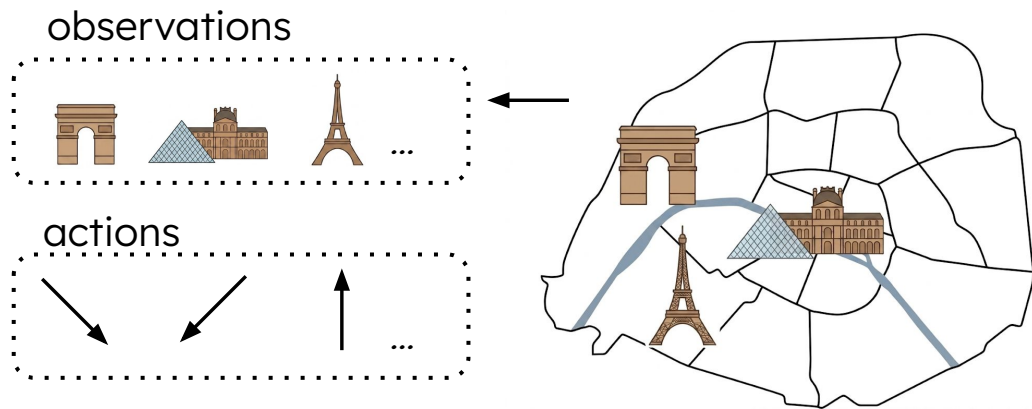
Eiffel Tower → Louvre → Arc de Triomphe



Fundamental question: How to robustly represent a dynamic world?



Clone-Structured Causal Graphs learn robust representations

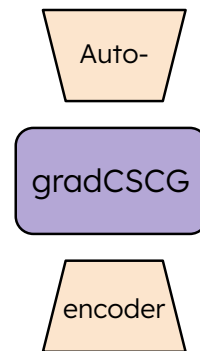


connections
reflect
environment

but

- 1) static environments
- 2) rusty architecture

preparatory work
gradient-based CSCG



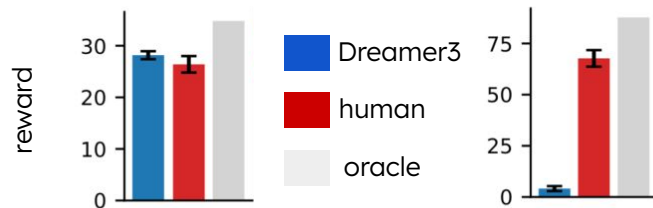
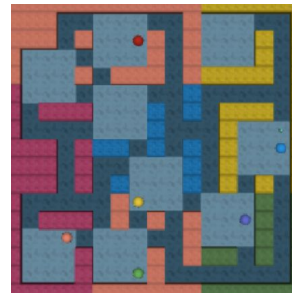
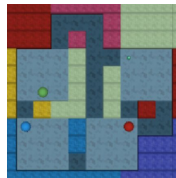
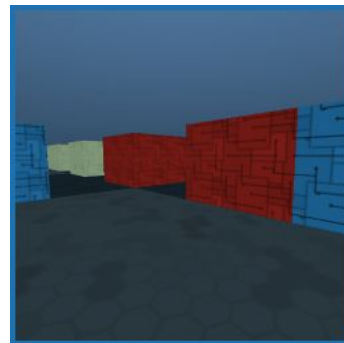
key innovation:
modularity



Dreamer3 leads across tasks ... but fails in Memory Maze.



Hafner et al., Nature (2025)



Pasukonis et al., arXiv (2022)

Objectives & Team

PhD 1

1) **Understand** why representations emerge?

PI

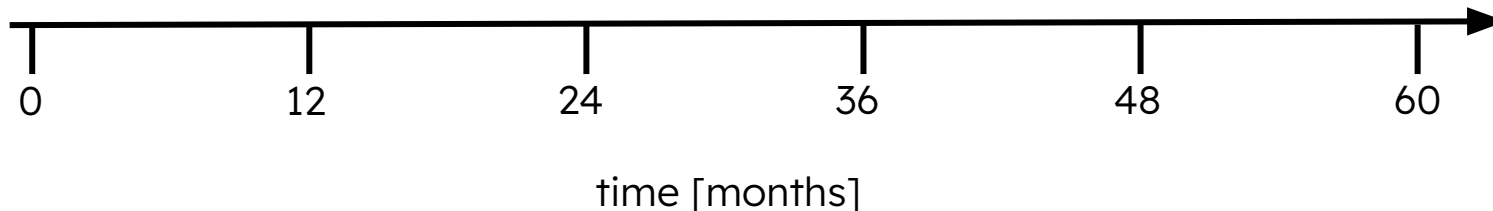
set-up

2) **Build** agents that act in dynamic environments

wrap-up

PhD 2

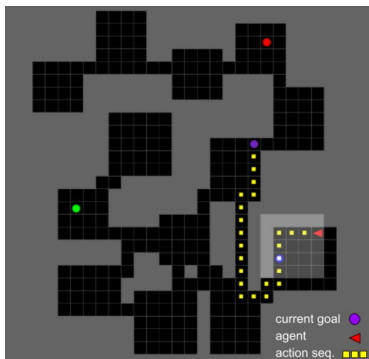
3) **Validate** against hippocampal recordings



1) **Understand** - why do robust representations (not) emerge?

empirical

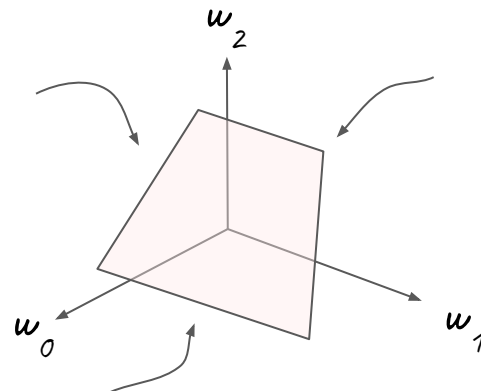
Memory Maze testbench



How are representations related to performance?

analytical

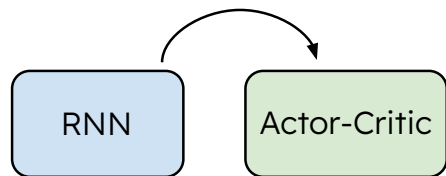
Representation learning as dynamical system



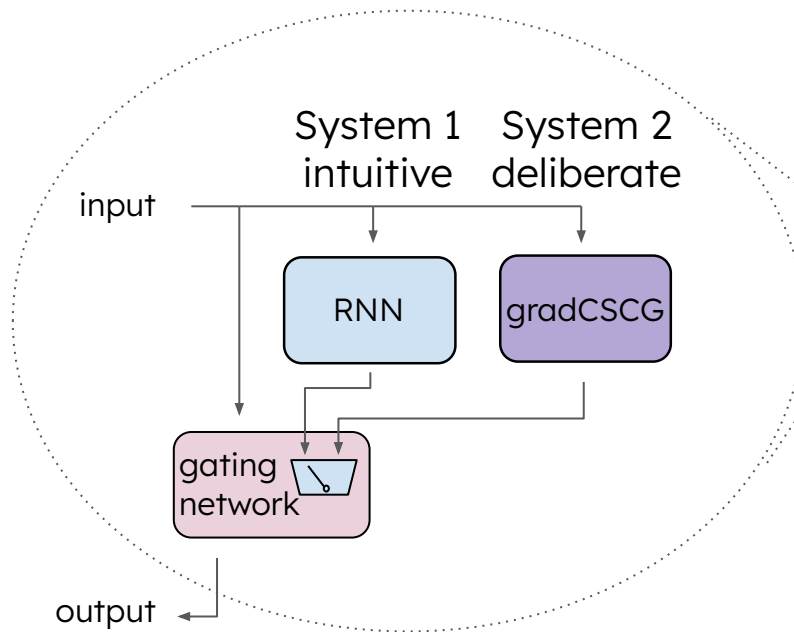
A mathematical microscope for an a-priori understanding

2) **Build** - agents that act in dynamic environments

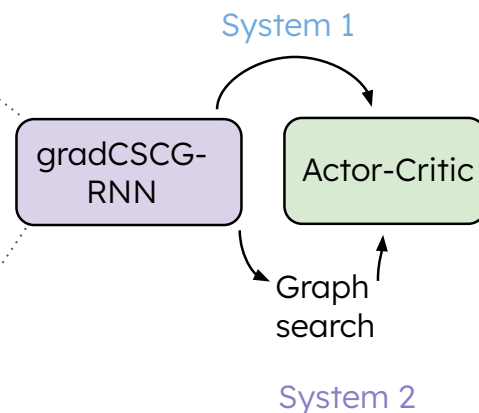
standard Dreamer



thinking fast and slow

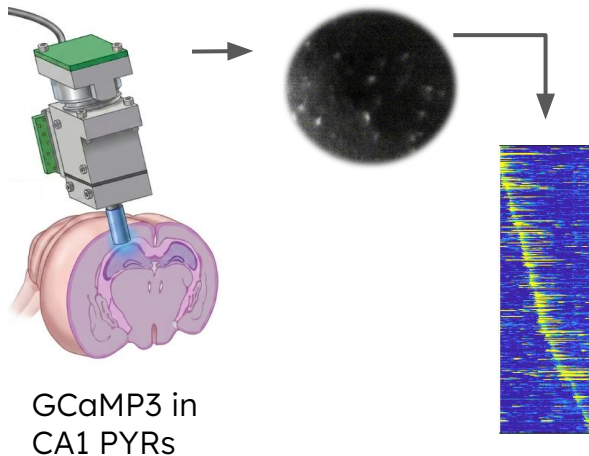


dual process in action



Massive CA1 recordings in dynamic environment

miniscope ~ 200-500 cells each



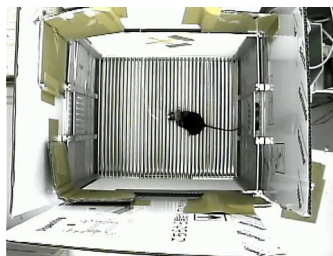
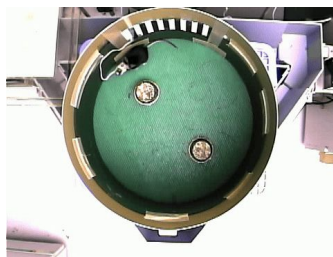
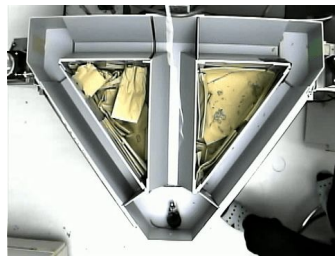
M	T	W	T	F	S	S
X	X	X	X	X		
X	X	X	X	X		
X	X	X	X	X		
X	X	X				
	X	X				

recordings on 21
of 38 days

>7h/animal

25 animals recorded

data transferred



Unpublished data by
Tom McHugh (RIKEN)



arenas

linear track, left/right



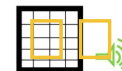
open fields + objects



elevated plus maze



trace fear conditioning

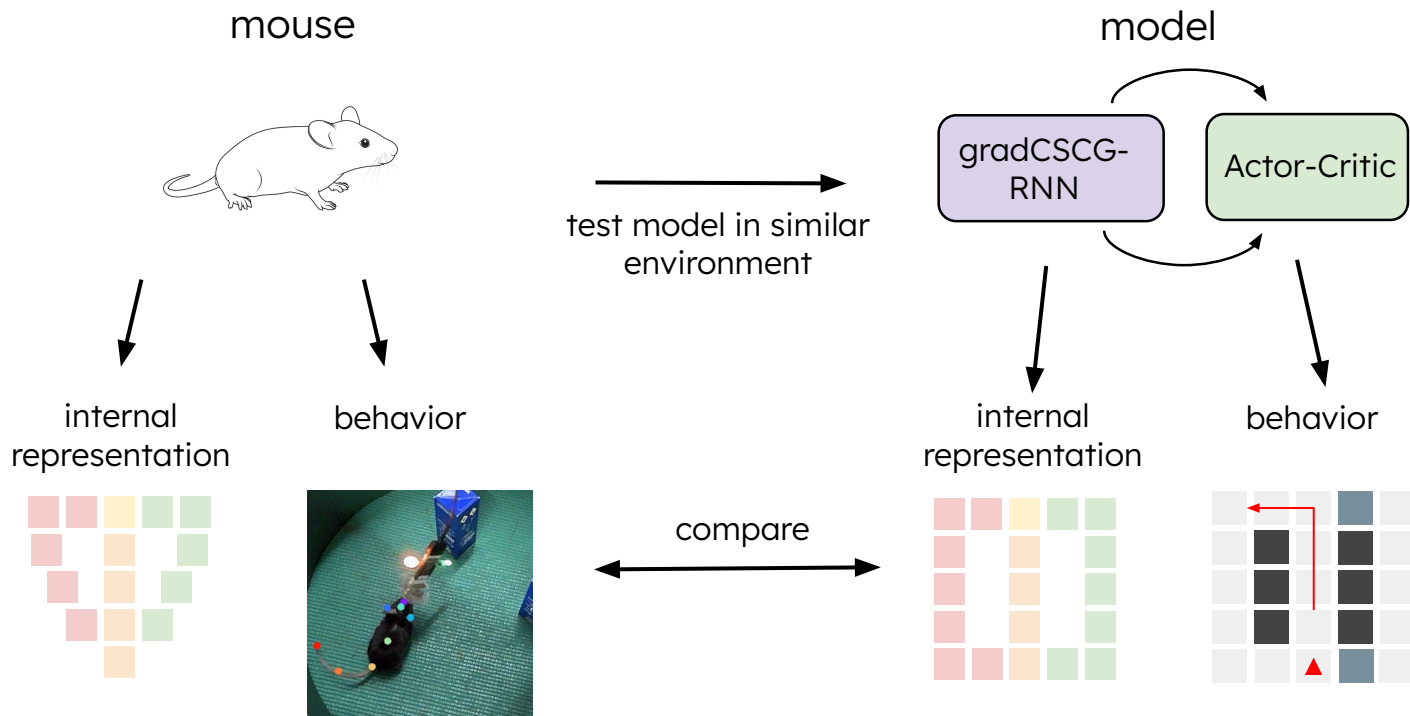


tone associations



11 different tasks/contexts in total

3) **Validate** model against hippocampal data



Do internal representations explain behavior?

Qualifications

documented expertise in key domains

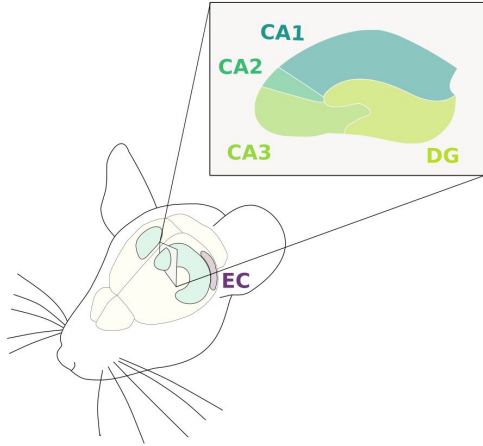
hippocampus	Lehr, ..., Stöber , Hippocampus. (2023)	Stöber et al., PLoS Comput. Biol. (2025)	Stöber et al., Hippocampus (2020)
	Alexander, ..., Stöber , et al., J. Neuro. (2025)	Pochinok, Stöber , et al., Nat. Commun. (2024)	Lehr & Stöber , PNAS (2021) Lehr, ..., Stöber , Neurosci. Biobehav. Rev. (2021)
other	Buccino, Stöber , et al., BIOCAS (2016)	Lepperød, Stöber , et al., PLoS Comput. Biol. (2023)	Stöber et al., Commun. Biol. (2023)
		Vieth, Stöber , Triesch, Front. Neuroinform. (2021)	
data analysis		artificial neural networks	conceptual

leadership & independence

- 6 articles without PhD supervisor
- 3 articles as last author
- supervision: 1 PhD, 4 MSc, 5 assistants, 12 interns
- >600k Euro raised since PhD
- independent fellowships (2)
- peer-recognition:
 - Johanna Quandt Young Academy, Frankfurt
 - Young Academy of Norway

Impact: From biological insight to artificial foresight

neuroscience



reinterpret hippocampal circuit

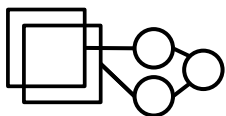
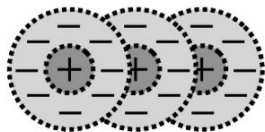
artificial intelligence



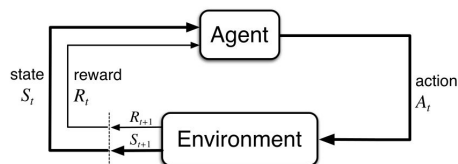
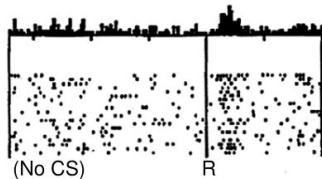
reliable and data-efficient algorithms

WoM-BaM is at the center of the next NeuroAI frontier

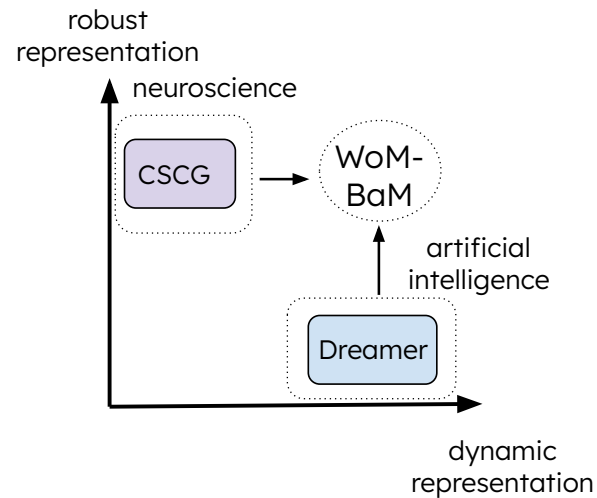
Visual system
and
ConvNets



Dopamine
and
Reinforcement Learning



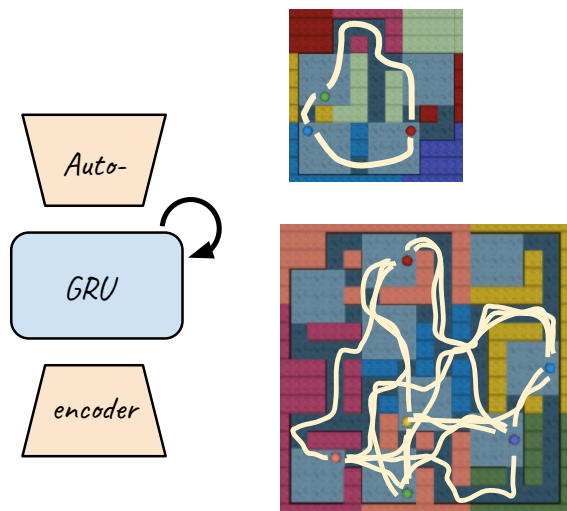
Hippocampus
and
World Models



Thank you!

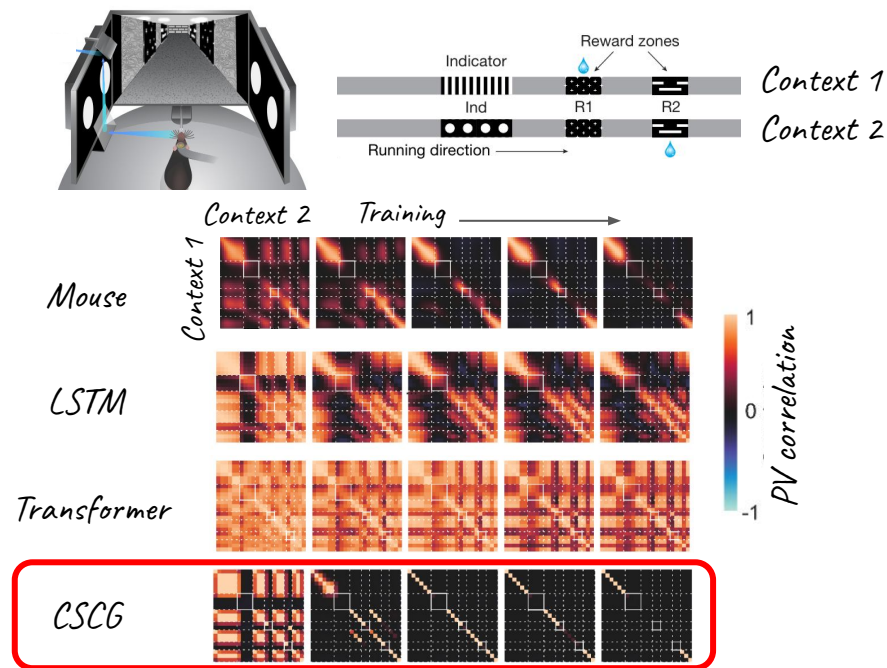
Backup slides

Dreamer may have flawed representations

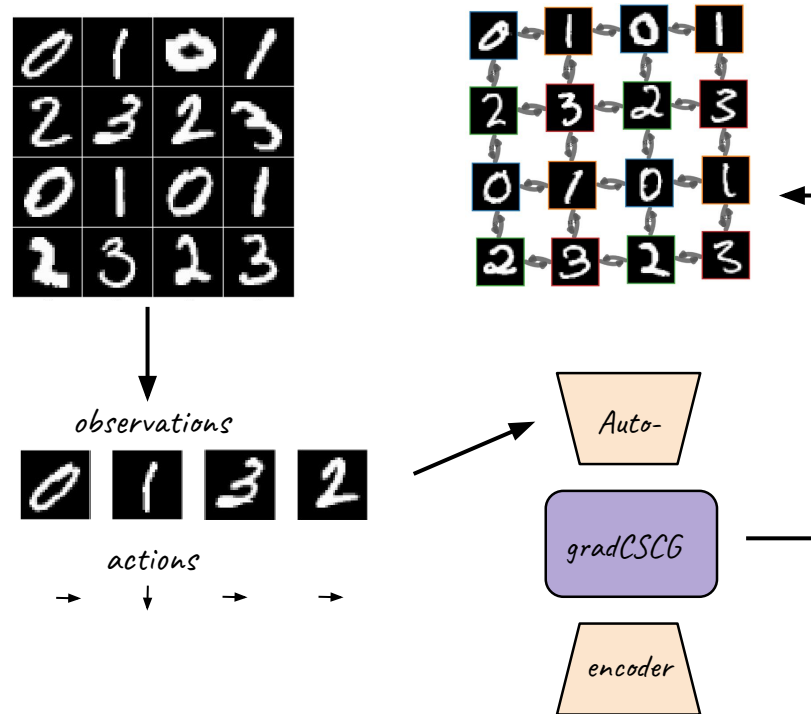


$$n_{\text{paths}} \sim n_{\text{objects}}^2$$

Lack of orthogonalization in RNN and Transformer architectures



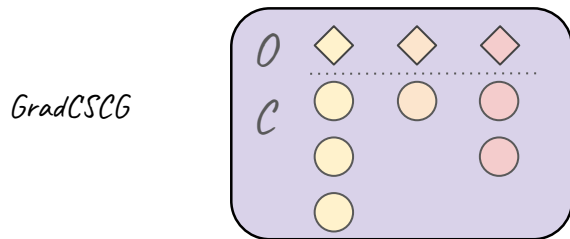
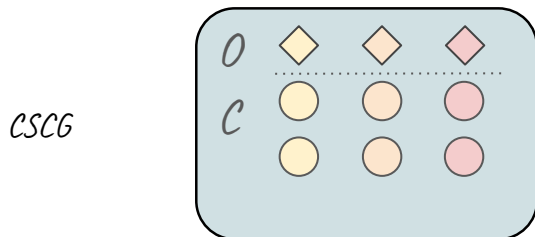
Fast and modular CSCG with a gradient-based reimplementation



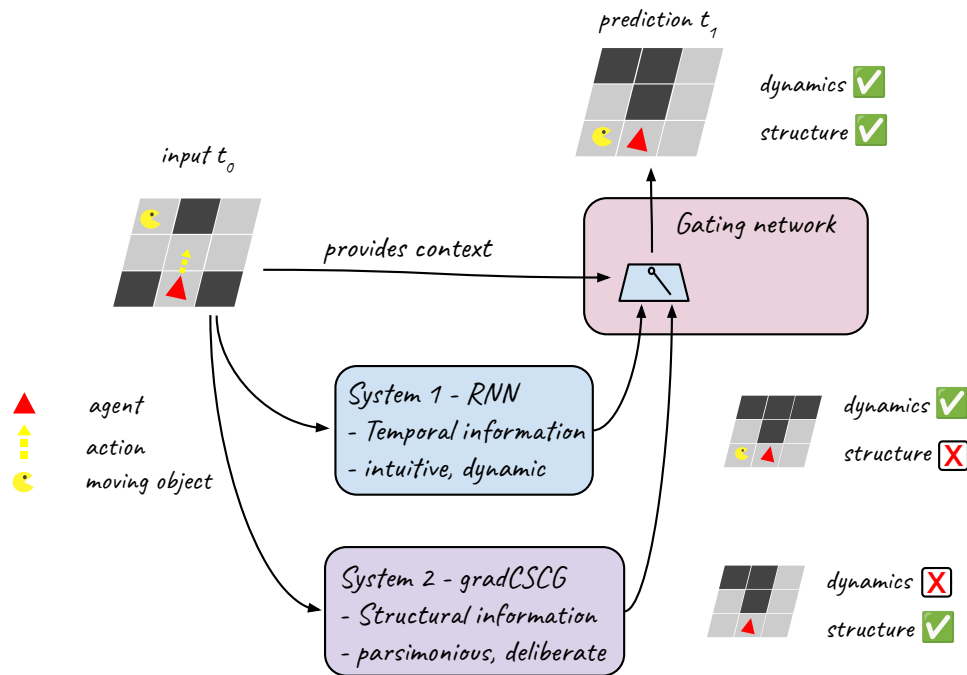
Obj. 2: From gradCSCG to a brain-inspired hybrid planning architecture

WP2.1: Improve scaling

- problem: clones are pre-initialised - inefficient!
- solution: sparsity regularizer on emission matrix
- speed: mask inactive clones

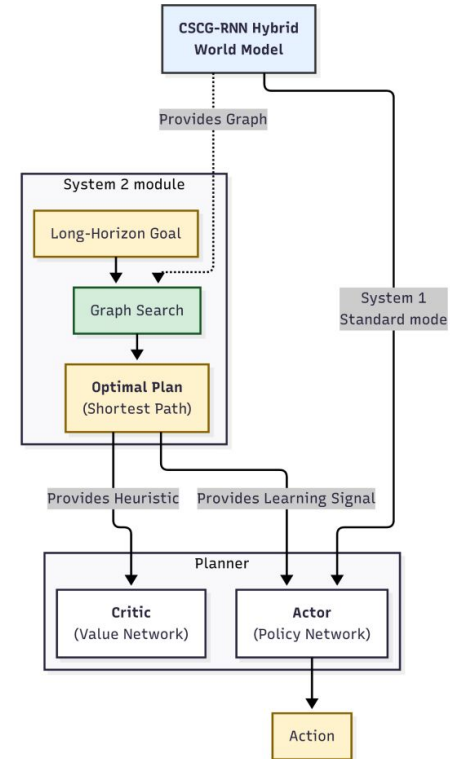
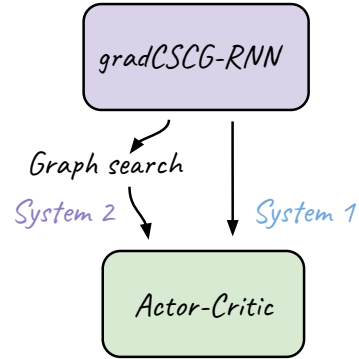
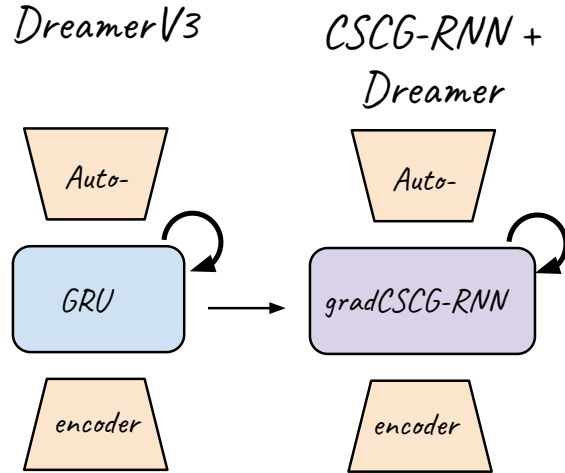


WP2.2: Combine gradCSCG with RNN



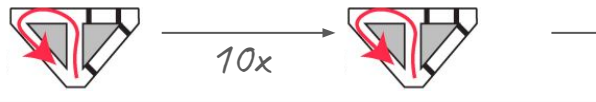
Obj. 2: From gradCSCG to a brain-inspired hybrid planning architecture

WP2.3: Graph-augmented planning



Combining arenas
and stimuli allows to
address
fundamental
questions

formation dynamics?



predictability?



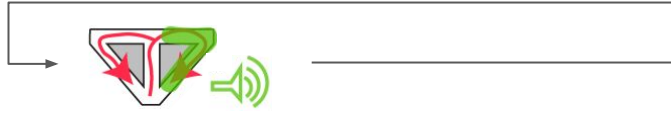
map extension or reuse?



saliency cue?



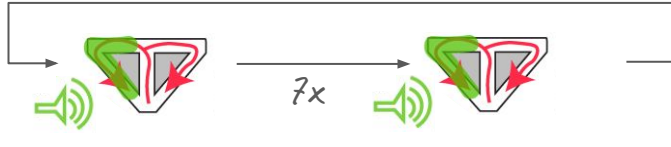
preference?



anticipation and reinforcement?



context reversal and habituation?



long term retrieval?



11 tasks/contexts

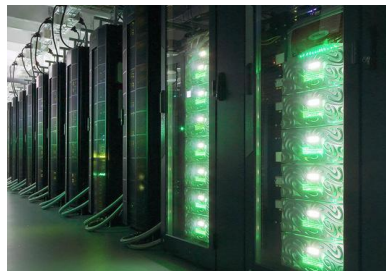
1. Linear Track Exploration (Left)
2. Linear Track Exploration (Right)
3. Open Field Exploration
4. Elevated Plus Maze
5. Object Location Recognition
6. Novel Object Recognition
7. Trace Fear Conditioning (TFC) Learning
8. TFC Retrieval Test
9. Linear Track with Tone Association
10. TFC Learning Boost Session
11. Long-Term TFC Retrieval Test

Team + infrastructure + collaborators

Institute



Compute



Collaborators

*Tom
McHugh*



data

*Dileep
George*



CSGC

*Alexander
Ecker*



*deep
learning*

*Andrew
Lehr*



*neuronal
geometry*

Objectives and work packages		Activity lead	Year 1 2027	Year 2 2028	Year 3 2029	Year 4 2030	Year 5 2031
Obj. 1 MiniGrid memory maze as testbench for internal representations							
WP1.1	Implementing the MiniGrid Memory maze	Ph.D. 1		M1.1			
WP1.2	Comparing models in MiniGrid Memory maze	Ph.D. 1			M1.2		
WP1.3	Theoretical analysis via Dynamical Systems Framework	Ph.D. 1				M1.3	
Obj. 2 Developing a brain-inspired hybrid planning architecture							
WP2.1	Scalable and Dynamic CSCG through Learnable Components	PI		M2.1			
WP2.2	Hybrid Architectures for Flexible Temporal Integration	PI			M2.2		
WP2.3	Graph-Augmented Planning in Model-Based RL	PI				M2.3	
Obj. 3 Testing and explaining hippocampal activity in dynamic environments							
WP3.1	Preprocessing and Activity Extraction from Miniscope Data	Ph.D. 2		M3.1			
WP3.2	Characterizing Hippocampal Representations ...	Ph.D. 2			M3.2		
WP3.3	Normative models of hippocampal activity in complex env.	Ph.D. 2				M3.3	
Project management and dissemination							
	Continuous risk and quality management	PI & Team					
	Onboarding and technical setup	PI					
	Consolidation of findings	PI					
	Offboarding and documentation	PI & Team					
	Dissemination	PI & Team					